

a. Monocot



b. Eudicot

Figure 23.21 Flower diversity. Regardless of size and shape, flowers, such as this (**a**) monocot and (**b**) eudicot, share certain features.
(a): Ed Reschke/Stone/Getty Images; (b): Steven P. Lynch

structure, a saclike container called the anther. Pollen grains develop from microspores produced within the anther.

- At the very center of a flower is the **carpel,** a vaselike structure with three major regions: the stigma, an enlarged, sticky knob; the style, a slender stalk; and the **ovary,** an enlarged base that encloses one or more ovules. The ovule becomes the seed, and the ovary becomes the fruit. Fruit is often instrumental in the distribution of seeds.

Note that not all flowers have all these parts (Table 23.3). A flower is said to be *complete* if it has all four parts; otherwise, it is *incomplete.*

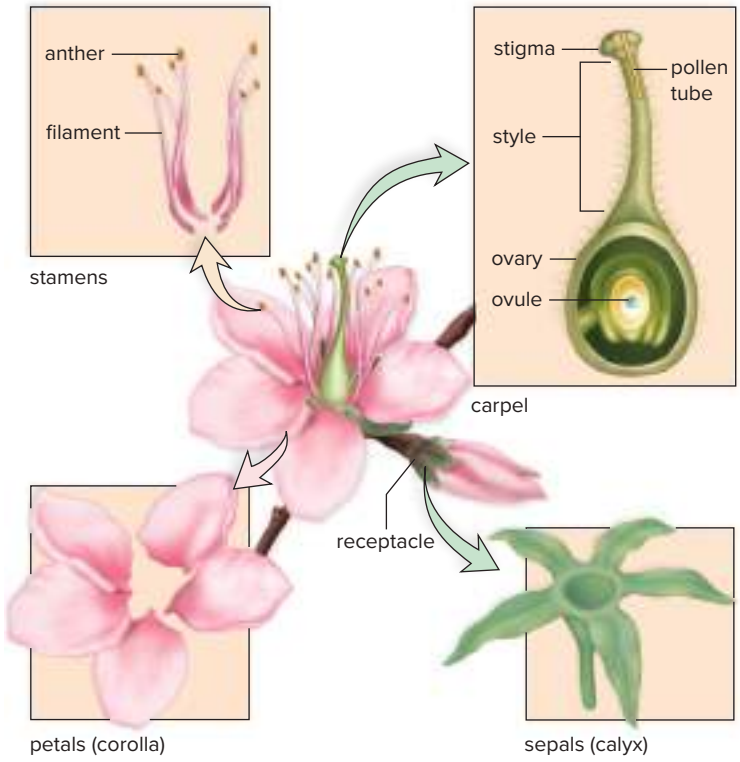


Figure 23.22 Generalized flower. A flower has four main parts: sepals, petals, stamens, and carpels. A stamen has an anther and filament. A carpel has a stigma, style, and ovary. An ovary contains ovules.

Table 23.3 Other Flower Terminology

Term	Description of Flower
Complete	All four parts (sepals, petals, stamens, and carpels) present
Incomplete	Lacks one or more of the four parts
Perfect	Has both stamens and (a) carpel(s)
Imperfect	Has stamens or (a) carpel(s), but not both
Inflorescence	A cluster of flowers
Composite	Appears to be a single flower but consists of a group of tiny flowers

Flowering Plant Life Cycle

Figure 23.23 depicts the life cycle of a typical flowering plant. Like the gymnosperms, flowering plants are heterosporous, producing two types of spores. A megaspore located in an ovule within an ovary of a carpel develops into an egg-bearing female gametophyte, called the embryo sac. In most angiosperms, the embryo sac has seven cells; one of these is an egg, and another contains two polar nuclei. They are called the polar nuclei because they came from opposite ends of the embryo sac.

Microspores, produced within anthers, become pollen grains that, when mature, are male gametophytes with sperm. The mature male gametophyte consists of only three cells: the tube cell and two sperm cells.

During pollination, pollen is transported from the anther to the stigma of a carpel. Here, the tube cell produces a pollen tube that

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